# 3.4 Derivatives of Inverse Trigonometric Functions — teacher key

**Question:** How fast does an angle change when its sine, or its tangent, changes?

## Self-paced lesson

No clock, no trimming, no skipping, no end mark, no stamp. Every tab is required in order; progress saves on the device so a student can stop and finish the next day.

## Warm-up review (3 easy items)

- The derivative of $\sin(5x)$ is... **5cos(5x).** Chain rule: the outside gives $\cos(5x)$, and the inside $5x$ brings a factor 5.
- If $\sin y = x$, then differentiating both sides with respect to $x$ gives... **cos y · dy/dx = 1.** Every $y$ term gets a factor $\frac{dy}{dx}$. The derivative of $\sin y$ is $\cos y \cdot \frac{dy}{dx}$, and the derivative of $x$ is 1.
- A right triangle has hypotenuse 5 and one leg 3. The other leg is... **4.** Pythagorean theorem: $3^2 + 4^2 = 5^2$, so the other leg is 4.

## Explore stations

Four hands-on stations (X1, X2, X3, X4). Station answers are in the teacher view of the lesson page (add `#key` to its address).

## Explore: what did you see

- In Station 3, the slope of arctan was largest at $x =$... **0.** The slope was 1 at $x = 0$ and got smaller as $x$ moved away.
- In Station 4, at the same $x$, the slope of arccos compared with the slope of arcsin was... **the opposite sign and the same size.** The two slopes were 1.25 and $-1.25$, then 1.67 and $-1.67$: same size, opposite sign.

## Mid-lesson check (Huddle tab)

- For $y = \arctan(4x)$, the derivative $y'$ is... **4 / (1 + 16x²).** Chain rule: $u = 4x$, $u' = 4$, and $u^2 = 16x^2$. So $y' = \dfrac{4}{1 + 16x^2}$.
- For $y = \arcsin(2x)$, the expression under the square root is... **1 − 4x².** Square the whole inside: $(2x)^2 = 4x^2$, not $2x^2$.

## Targets

- I can say what arcsin, arccos and arctan mean.
- I can find the derivative of an inverse trig function, with the chain rule.
- I can find a slope or a tangent line for an inverse trig curve.

### Predictions

- Sine has a flat tangent at its peak. Mirror that piece across $y = x$. At $x = 1$, the arcsin curve has a tangent that is... **vertical.** Rise and run swap in a mirror. A flat tangent becomes a vertical tangent, so the slope of arcsin grows without bound near $x = 1$.
- The graph of arctan levels off for large $x$. As $x$ gets larger, its slope gets... **closer to 0.** A curve that levels off has a tangent that gets flatter and flatter, so the slope heads toward 0.

## Huddle script (answers in green)

### Definitions

- **arcsin x:** The **angle** whose sine is x. It is not 1/sin x.
- **Inverse rule:** If y = arcsin x, then sin y = **x**. Differentiate both sides to find dy/dx.
- **The three rules:** arcsin: $1/\sqrt{(1 - x^2)}$. arccos: $-1/\sqrt{(1 - x^2)}$. arctan: $1/(1 + \mathbf{x^2})$.
- **Chain rule:** Put u in for x, then multiply by **u?**. Square the **whole inside**.

### Worked example

Differentiate y = arcsin(3x).

- The inside is u = **3x**
- Its derivative is u? = **3**
- Square the whole inside: $u^2$ = **$9x^2$**
- So dy/dx = 3 / $\sqrt{(1 - 9x^2)}$

### Label the triangle

1. The longest side has length **1**.
2. The side opposite y has length **x**.
3. The third side has length **$\sqrt{(1 - x^2)}$**.

## Every level, with answers

### Differentiate

**Level 1.** Differentiate.
**Answer:** $\frac{-1}{\sqrt{1-x^2}}$ Use the arccos rule directly.

**Level 2.** Differentiate.
**Answer:** $\frac{5}{\sqrt{1-(5x)^2}}$ The inside is $u = 5x$ and $u' = 5$. Put them in the arcsin rule.

**Level 3.** Differentiate.
**Answer:** $\frac{2x}{1+(x^2)^2}$ The inside is $u = x^2$ and $u' = 2x$. Put them in the arctan rule.

**Level 4.** Differentiate.
**Answer:** $\frac{\frac{-1}{x^2}}{1+\left(\frac{1}{x}\right)^2}$ The inside is $u = \dfrac{1}{x}$ and $u' = -\dfrac{1}{x^2}$. Put them in the arctan rule.

## Slope at a point

**Level 1.** $y = \arctan x$. Find $\dfrac{dy}{dx}$ at $x = 3$.
**Answer: 0.1** $u(3) = 3$ and $u'(3) = 1$. Put them in the arctan rule: the slope is 0.1.

**Level 2.** $y = \arcsin(5x)$. Find $\dfrac{dy}{dx}$ at $x = 0.12$.
**Answer: 6.25** $u(0.12) = 0.6$ and $u'(0.12) = 5$. Put them in the arcsin rule: the slope is 6.25.

**Level 3.** $y = \arcsin(0.6e^x)$. Find $\dfrac{dy}{dx}$ at $x = 0$.
**Answer: 0.75** $u(0) = 0.6$ and $u'(0) = 0.6$. Put them in the arcsin rule: the slope is 0.75.

**Level 4.** $y = \arcsin(0.6e^x)$. Find $\dfrac{dy}{dx}$ at $x = 0$.
**Answer: 0.75** $u(0) = 0.6$ and $u'(0) = 0.6$. Put them in the arcsin rule: the slope is 0.75.

## Spot the right rule

**Level 1.** Which expression is $\dfrac{dy}{dx}$ for $y = \arccos x$?
(A) $1 \,/\, (1 + x^2)$ (B) $-1 \,/\, \sqrt{(1 - x)}$ (C) $1 \,/\, \sqrt{(1 - x^2)}$ **(D) $-1 \,/\, \sqrt{(1 - x^2)}$**
**Answer: D** The inside is $u = x$, so $u' = 1$ and $u^2 = x^2$. The arccos rule gives $-1 \,/\, \sqrt{(1 - x^2)}$.

**Level 2.** Which expression is $\dfrac{dy}{dx}$ for $y = \arcsin(5x)$?
**(A) $5 \,/\, \sqrt{(1 - 25x^2)}$** (B) $5 \,/\, (1 + 25x^2)$ (C) $5 \,/\, \sqrt{(1 - 5x)}$ (D) $1 \,/\, \sqrt{(1 - 25x^2)}$
**Answer: A** The inside is $u = 5x$, so $u' = 5$ and $u^2 = 25x^2$. The arcsin rule gives $5 \,/\, \sqrt{(1 - 25x^2)}$.

**Level 3.** Which expression is $\dfrac{dy}{dx}$ for $y = \arctan(x^2)$?
(A) $1 \,/\, (1 + x?)$ (B) $2x \,/\, (1 + x^2)$ (C) $2x \,/\, \sqrt{(1 - x?)}$ **(D) $2x \,/\, (1 + x?)$**
**Answer: D** The inside is $u = x^2$, so $u' = 2x$ and $u^2 = x^4$. The arctan rule gives $2x \,/\, (1 + x?)$.

**Level 4.** Which expression is $\dfrac{dy}{dx}$ for $y = \arctan(e^x)$?
(A) $e? \,/\, (1 + e?)$ (B) $e? \,/\, \sqrt{(1 - e^2?)}$ **(C) $e? \,/\, (1 + e^2?)$** (D) $1 \,/\, (1 + e^2?)$
**Answer: C** The inside is $u = e^x$, so $u' = e^x$ and $u^2 = e^{2x}$. The arctan rule gives $e? \,/\, (1 + e^2?)$.

## Tangent line

**Level 1.** Write the tangent line to $y = \arctan x$ at $x = 0$.
**Answer:** $1\,(x - 0) + 0$ The point is $(0, 0)$ and the slope is 1. So $y = 1(x - 0) + 0$.

**Level 2.** Write the tangent line to $y = \arctan(4x)$ at $x = 0.25$.
**Answer:** $2\,(x - 0.25) + \frac{\pi}{4}$ The point is $(0.25, \frac{\pi}{4})$ and the slope is 2. So $y = 2(x - 0.25) + \frac{\pi}{4}$.

**Level 3.** Write the tangent line to $y = \arctan\left(\dfrac{1}{x}\right)$ at $x = 1$.
**Answer:** $(-0.5)\,(x - 1) + \frac{\pi}{4}$ The point is $(1, \frac{\pi}{4})$ and the slope is $-0.5$. So $y = -0.5(x - 1) + \frac{\pi}{4}$.

**Level 4.** Write the tangent line to $y = \arctan\left(\dfrac{1}{x}\right)$ at $x = 1$.
**Answer:** $(-0.5)\,(x - 1) + \frac{\pi}{4}$ The point is $(1, \frac{\pi}{4})$ and the slope is $-0.5$. So $y = -0.5(x - 1) + \frac{\pi}{4}$.

## Common misconceptions

| Students say | Say instead |
|---|---|
| $\sin^{-1} x$ means $1 / \sin x$. | It means the inverse function arcsin x. The notation looks like a power but is not one. |
| arcsin and arccos have the same derivative. | They differ by a minus sign. arccos is the negative of arcsin. |
| The derivative of arctan x has a square root. | Only arcsin and arccos do. arctan is $1 / (1 + x^2)$. |
| Forgetting u?: $d/dx \arctan(4x) = 1/(1 + 16x^2)$. | Multiply by the derivative of the inside: $4 / (1 + 16x^2)$. |
| Squaring only the x: $(3x)^2 = 3x^2$. | Square the whole inside: $(3x)^2 = 9x^2$. |